Optical Character Recognition for Pre-Digital Historical Documents using Large Language Models

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***Abstract*—Multi-modal large language models (LLMs) with vision capabilities have been shown to be a promising and effective approach to extracting text from images or optical character recognition (OCR). However, text extraction using OCR is challenging when applied to scanned pre-digital historical documents. Factors such as poor scan quality due to the age of the documents, various levels of fading, scan skew (left or right), and possible abnormal or poor background-to-text contrast can contribute to incorrect OCR results. Performing OCR on scanned pre-digital historical documents makes them machine-readable, thus enabling computational analysis and preservation. This can lead to a better understanding of the past, especially for significant events and time periods. Given the text extraction capabilities of LLMs with vision, we posit that they are a viable option for performing robust OCR on scanned pre-digital historical documents. We chose a set of foundational and capable OCR technologies to compare to LLMs with vision. To accomplish this, we curated a ground truth dataset comprising scanned pre-digital historical documents from the early twentieth century for comparing the chosen OCR technologies. Our experiments showed that LLMs with vision, specifically Mistral AI's Mistral-Small-3.1-24B-Instruct-2503 model and Allen AI's olmOCR-7B-0225-preview, are able to perform OCR on our dataset very well. We evaluated using the character error rate (CER), BLEU score, multiple ROUGE scores, and the Normalized Levenshtein Distance (NLD) for each OCR technology. Mistral had better results but the potential for large error (2 cases out of 359 from our dataset), while olmOCR performed almost as well but was more consistent in mitigating high error. These results support the use LLMs with vision to perform OCR on scans of pre-digital historical documents with challenging characteristics.**

***Index Terms*—Vision Large Language Models, LLM, Optical Character Recognition, OCR, Historical Documents**

I. Introduction

Optical character recognition (OCR) is well recognized for its ability to convert documents from print to digital formats with ever-increasing improvements to accuracy and efficiency. For example, OCR plays a pivotal role in libraries for archiving and making available large collections of physical documents. Currently, the usefulness of OCR is limited by the quality of the digitized content, which depends greatly on the quality of the scanned document. Our work focuses on extracting text from pre-digital historical documents from the early twentieth century, as described in detail in Section III-A. Scanned images from such collections can often be of low quality due to various levels of fading, scan skew (left or right), and text-to-background contrast. The latter can come from scanning the negative of a document image, which is common practice for archival preservation [1]. A scan could include one or more of these characteristics, making the OCR of such documents a challenging task.

To address challenges that are specific to scanned, typed, pre-digital historical documents, we investigate recently developed multimodal large language models (LLMs) with vision capabilities.

OCR technologies differ but often use a similar step-wise procedure. Generally, these include pre-processing images to optimize text detection (e.g., skew correction and color filtering), detection of where the text is in an image, and text recognition on detected text region(s) [2], [3]. One may not know the exact condition of the scanned documents ahead of time; thus, deciding upon a specific pre-processing step for the scans may not be feasible, especially when dealing with a large number of documents. Hence, we also focus on applying OCR to the original scanned document images without any pre-processing.

To make the results widely beneficial to the research community, we focused on open-access LLMs available to researchers, i.e., LLMs with no paywall. We also focus on the smaller parameter version of each LLM to show the effectiveness of LLMs that can be run on cost-effective hardware, demonstrating real-world applications with quality results without the need for a potentially expensive computing system. The OCR technologies compared are listed in Table I.

Our contributions are three-fold. First, we curated a new dataset[1] from scanned pre-digital historical documents with challenging characteristics with provided transcriptions. This provides the research community with a ground truth dataset to test their own OCR technology. Second, we performed extensive testing mid-June to early July 2025 with widely used OCR technologies. This provides a benchmark for the development, refinement, and training of OCR technologies for scanned documents with the previously described challenging characteristics. The date range of our experiments is given as new models are released often, hence, the reader knows

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TABLE I: OCR technologies used.

OCR Technology	Version	GPU
pytesseract	0.3.13 [5]	False
docTR	0.12.0 [6]	Both
surya	0.14.6 [7]	Both
Llama 3.2	Llama-3.2-11B-Vision-Instruct [8]	True
olmOCR	olmOCR-7B-0225-preview [9]	True
Mistral	Mistral-Small-3.1-24B-Instruct-2503 [10]	True
Pixtral	Pixtral-12B-2409 [11]	True
Llava 1.6	llava-v1.6-vicuna-13b-hf [12]	True
Phi-4	Phi-4-multimodal-instruct [13]	True

the cut-off time of the OCR technologies available. Lastly, we showcase that LLMs with vision are able to perform OCR on scanned pre-digital historical documents, especially without the need for image pre-processing. This paves the way for text analysis of the extracted text making it more possible to explore information that originally had very limited computational access.

The impact of LLMs with vision to perform OCR is far-reaching. They can greatly enhance text extraction for researchers in the humanities and for library collections by allowing them to perform text analysis and preservation on otherwise computationally inaccessible analog information without the burden of manual transcription. This is especially true for large document collections, such as newspapers. We believe that LLMs with vision can enable researchers to uncover buried historical information that has been captured so far by analog means with greater ease. For example, LLMs can give researchers access to the text content of documents related to the two world wars [4]. Furthermore, the creation of a single OCR pipeline that can handle paper aging, ink fading, scan skew, and contrast without the need for pre-processing can greatly increase the volume of documents that can have text extracted with improved accuracy.

II. Background

We present several open-source OCR technologies that we use in this work. The choice of these OCR technologies was based on cross-referencing several articles to find the latest suggestions of the best OCR technologies available, both traditional and LLMs [14], [15], [16], [17]. A comprehensive list of all OCR technologies and their corresponding versions used in the experiments presented in this paper is provided in Table I.

EasyOCR [18] was tested, but it performed poorly on our dataset. Thus, it wasn't chosen for our experiments, discussed in Section III-B. **Tesseract** [19] is an OCR engine initially developed by Hewlett-Packard in 2004 and developed and maintained by Google since 2006, It is openly available and supports images in PNG, JPEG, and TIFF formats. The latest version uses long short-term memory (LSTM) for line and character recognition. **Pytesseract** [5] is a Python wrapper for Tesseract and can be used as a stand-alone invocation script for Tesseract. **docTR** [6], [20] is an end-to-end OCR library designed to support text detection and recognition through various deep-learning models. The library provides users with a wide variety of options for text detection and recognition from scanned documents, PDFs, and images, particularly documents with challenging layouts. We choose the Pytorch-based `fast_base` model with 16.3 million (M) parameters for text detection and `parseq` with 15.8M for text recognition. Both models were chosen based on their performance on benchmark datasets (FUNSD and CORD) and the number of parameters in the base models as reported in the official documentation. **Surya** [7] is an OCR toolkit supporting line-level detection, layout analysis, table recognition, and LaTeX OCR. We use the default text detection (vikp/surya_det3) and recognition models (vikp/surya_rec2).

LLM OCR Technologies: LLMs specifically the vision-enabled multi-modal ones, have had success in performing OCR on different kinds of documents. We use HuggingFace's [21] PyTorch transformers library and LLMs hosted by them. Meta's `Llama-3.2 Vision` [22] is a vision-enabled LLM that has been pre-trained and contains 11 billion (B) or 90B parameters. The data used for pre-training comprised 6B image and text pairs and around 3M synthetically generated examples used for instruction tuning. The models are multi-modal in nature, which means they support images alongside text as the input. **Llama 3.2** has been shown to be capable of performing several tasks [23], such as image reasoning and extracting text from images. We use `Llama-3.2-11B-Vision-Instruct` [8] model for OCR purposes. **olmOCR** [24] is an LLM-based OCR tool built on top of `Qwen2-VL-7B-Instruct` [25], designed by Allen AI for converting PDFs and other documents into plain text. The model was trained using academic papers, technical documentation, and PDFs with diverse properties. including graphics, handwritten text, and poor-quality scans. We use `olmOCR-7B-0225-preview` [9], a 7B LLM. **Phi-4** [26] is a lightweight open multimodal foundation model by Microsoft capable of processing text, image, and audio inputs. We use `Phi-4-multimodal-instruct` [13], a 5.6B parameter model for our experiments. **Pixtral** [27] is multimodal LLM trained on image and text data that supports multiple images. We use `Pixtral-12B-2409` [11], which comprises a 12B parameter multimodal decoder and a 400M parameter vision encoder. **Mistral** [28] is a 128K context window supporting multi-modal LLM. The model has been shown to perform well on various multimodal benchmarks while maintaining an inference speed of 150 tokens per second. We use `Mistral-Small-3.1-24B-Instruct-2503` [10], a 24B parameter instruct model. **Llava** [12] is a visual instruction tuned LLM that connects a vision encoder and LLM (support for Vicuna-7B, Mistral-7B, and Hermes-Yi-34B available) for language and visual understanding. We use Llava with Vicuna as the LLM, specifically `llava-v1.6-vicuna-13b-hf` model with 13B parameters for our OCR experiments.

We must note that the choice of LLMs with vision was motivated by the nature of our dataset. The dataset is in English, which led us to choose widely used LLMs with vision

that support English. To find the best OCR technology (both LLM and non-LLM) that works on our dataset, we had to test a range of models that were trained on varied data and have different kinds of architectures.

Evaluation Metrics: We use several metrics to evaluate the quality of the results generated from the aforementioned models. **CER** or 'Character Error Rate' [29] is a metric that helps measure the percentage of incorrectly transcribed characters. CER takes into account the number of substitutions, deletions, and insertions when evaluating reference text quality. A CER of 0 means there is a perfect match between the reference and the machine-generated text. CER has been used to curate our ground truth dataset as well as evaluate OCR results. **ROUGE Score** or 'Recall-Oriented Understudy for Gisting Evaluation' [30] is a metric to measure the quality of generated text against ground truth or reference text. While ROUGE is popularly used for summarization and translation, it can also be used for transcription. For transcription tasks, ROUGE score can help evaluate the quality of machine transcription (generated text) compared to the human transcription (reference text). Different kinds of ROUGE metrics are: **ROUGE-N** measures the overlap of n-grams between the machine-generated transcribed text and the reference text. **ROUGE-L** measures the longest common subsequence between the machine-generated transcribed text and the reference text. The scores are between 0 and 1. A score of 1 depicts a perfect match between the machine-generated transcribed text and the reference text.

BLEU Score or 'Bilingual Evaluation Understudy' [31] is a metric that measures the similarity between machine-translated and reference translation text. Originally developed for evaluating machine translation, the score correlates well with human judgment. We use the BLEU score to compare the quality of the machine-generated transcribed text to a reference transcript. A score of 1 depicts a perfect match between the machine-generated transcribed text and the reference text. Normalized edit distance is often used to compare strings [32]. We use **NLD** or the 'Normalized Levenshtein Distance' as a metric to calculate the results of our OCR transcriptions. The formula uses the edit distance metric based on the Levenshtein distance [33], which measures the minimum number of single-character edits, i.e., insertions, deletions, and substitutions, needed to change one string into another. The normalized formula takes into account the number of edits between two strings divided by the length of the longest string among the two. This score ranges from 0 to 1. A lower edit distance score signifies a better match between the machine-generated transcribed text and the reference text.

III. Methodology

A. Dataset

Our dataset comprised a collection of early twentieth-century legal real estate documents found in the United States that focus on racial restrictions of occupation and ownership. This dataset is a subset of documents from the Chicago Covenants project [34]. To create baselines for testing the performance of LLMs with vision on this dataset, we curated a data subset as described in this section. The current analog form of the collection limits our ability to analyze and draw meaningful insights at scale without laborious close reading. The documents reflect different pre-determined templates used with slight variations throughout the years.

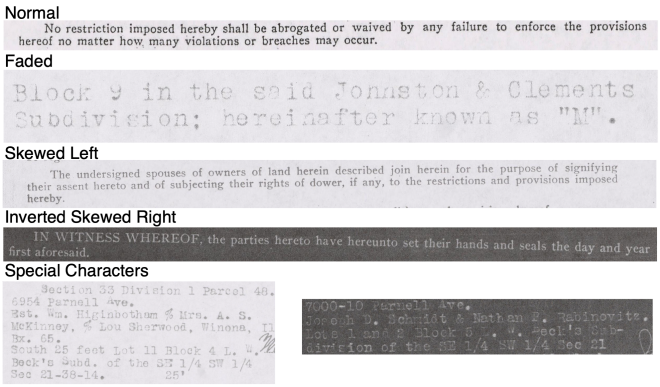


Fig. 1: Various Characteristics of the document scans.

This paper focuses on a subset of the collection that represents a gamut of challenging characteristics of scanned documents for OCR technologies. From the document collection, we identified four categories of document scan characteristics that can pose a challenge: *faded*, *skewed*, *inverted* background/text colors, and *special characters*. Fig. 1 provides some examples. For simplicity, we deemed that three (minus *special characters*) have sub-categories describing the different scan or document image types' features. *Faded* can be *weakly faded* or *very faded*, depending on the scan quality and the condition of the original document. *Skewed* can also be *weakly skewed* either to the left or the right or have a *very strong skew* angle (left or right). Lastly, some of the scans have an inverted black background with white text instead of black text on a white background. Another obstacle is that the documents have the potential to be any number of combinations of these scan conditions, such as *faded* and *skewed*. The *special characters* label resulted from observations of characters that are unique cases. These characters are measurement tick marks on numbers, fractions with a horizontal bar (e.g., $\frac{1}{2}$), / (division), #, and %.

We started with 30 real estate agreement documents ranging from one page to six pages (81 pages in total). Cropped clips were chosen based on variations of the challenging characteristics and different text lengths. The final set was 125 cropped clips. These clips comprise varying text lengths from one sentence to very detailed paragraphs. Character counts for the clips range from 76 to 1,350 with a median of 265. Word counts for the clips range from 13 to 232 with a median of 41. Since these were legal documents with formulaic structure and formatting, many sections had the same text with variations depending on year of creation. Hence, we relied on the mentioned challenging characteristics to differentiate text sections (even if the text itself was the same).

TABLE II: Counts of the different labeled categories broken down by original (orig.) and augmented (aug.) clips.

Label Name	Orig./Aug. Presence	Orig./Aug. Absence	Total Presence	Total Absence
Normal	18 / 0	107 / 234	18	341
Faded	43 / 72	82 / 162	115	244
Skewed	79 / 144	46 / 90	223	136
Inverted	22 / 126	103 / 108	148	211
Skewed Right	40 / 72	85 / 162	112	247
Skewed Left	39 / 72	86 / 162	111	248

Two researchers each took 15 real estate agreement documents from the original set of 30 and independently made clips based on the defined categories: *normal*, *faded*, *skewed*, *inverted*, and *special characters*. Between the two researchers, 178 clips were identified. Then, the clips and the decided-upon labels (described shortly) were reviewed manually, and if two or more clips had the same text *and* the same label, only one was kept, and the duplicate(s) were excluded. This process resulted in 125 unique clips, i.e., 70% of the original identified clips.

Each clip was then independently transcribed by two researchers, and the differences were identified programmatically using the CER metric. Corrections to each of the transcribers' transcriptions were also independently completed. This was done iteratively until each transcription matched, i.e., CER of 0. Non-trivial discrepancies were discussed until a consensus was reached. The final transcriptions, each pair with a CER of 0, were our gold standard or reference transcriptions for our evaluation, discussed in subsequent sections.

For determining the labels for each clip, given the four overarching labels, *faded*, *skewed*, *inverted*, and *special characters*, each clip was labeled independently by two researchers, and discrepancies were discussed until a consensus was reached. Some clips contained text that was clear, had good text and background contrast, and were not *skewed* or *inverted*, for which they were labeled as *normal*. Table II details the breakdown of our dataset given these categories. Thirteen of the 125 clips were found to include *special characters*.

Since the initial dataset, 125 clips, is on the smaller side, we applied data augmentation to expand not only the number of clips to work with but also the variety to ensure a more robust evaluation. In order to deterministically create augmented version of the clips, we applied augmentations to the 18 identified *normal* clips. This allowed us to start from a known base, *normal*, and then create versions of *faded* (weak and strong), *skewed* (weak and very skewed in both left and right directions), and *invert* all the above, including the original *normal* clips. For each *normal* clip, we were able to create 13 variations, leading to an additional 234 clips and giving us a total dataset of 359 clips.

This corpus was chosen because there was a need for text extraction for the project analyzing these historical real estate documents. Since these documents contained the previously described challenging characteristics for OCR technologies, this led us to explore the possibilities of realizing the goal of text extraction.

B. Experiments

Each of the core 125 clips from our dataset was chosen to represent only stand-alone paragraphs so as to focus purely on the text extraction and not layout and text formatting. These aspects are part of future work. Table I shows the different OCR technologies used, the software version used, and if they were GPU (Graphical Processing Unit) enabled.

For each OCR technology, we applied it to each scanned clip and recorded the extracted text verbatim. Since formatting was not a focus, any newlines in the resulting identified text were removed. Each of the following metrics was calculated and recorded for each OCR technology: CER, NLD, BLEU, ROUGE1, ROUGE2, and ROUGEL. We report the average for each of these metrics in Tables III and IV. For CER, we also add the "Top CER %" metric for each OCR technology. The "Top CER %" is the number of clips that had a CER below 2% divided by the total number of clips. Hence, it shows a percentage of how many clips are within the conventionally desired CER range [35], [36].

For the LLMs, we used a single prompt to ask the LLM to extract the text. The prompt used was: **'Perform text identification of all typed text and punctuation into a single text line verbatim. Provide no explanation and no paragraph numbers.'** This prompt was developed when working with Llama 3.2, the first LLM with vision tested. We went through several iterations of the prompt before settling on the final one. An important keyword we observed was the use of the term "identification" rather than "extraction" where the former showed better results. Another important keyword was "verbatim" which helped improve text extraction, as otherwise the model would use synonyms or paraphrase the sentence(s). We also needed to be specific about what to extract (e.g., text and punctuation) and in what format (single line, not in paragraph form). We also asked to not provide an explanation and no paragraph numbers as we observed Llama 3.2 would provide this information, which we did not desire as we were focused purely on the text. Even though we needed to be this specific, this shows support for how flexible LLMs can be, as they can follow specific directions well. We should also note that for the LLM OCR technologies, we needed to reset the GPU memory for each clip to overcome what we believe to be the LLM context getting "in the way". We observed that an LLM model might generate text for a clip that had some text cut-off (e.g., a highly *skewed* clip with partial text from another paragraph visible) and would fill in what it thought the cut-off text would be. Plus, since the documents are formulaic and share many phrases across documents, we believe the LLMs would "remember" certain phrasings or sentence pieces and generate text based on those "memories" instead of what is actually in a clip. Despite needing this reset, the performance speed did not appear to be affected.

Our experiments were completed on the same computing system configuration to minimize any potential differences caused by varying hardware setups: 2 Nvidia L40s GPUs,

each with 48 GB of video RAM. We observed that most of the OCR technologies could run on solely one L40s GPU, namely all but Mistral and Pixtral (depending on LLM parameters). Each LLM was run without being quantized. Given these resources, we used the smaller versions of each LLM. Comparatively, this system setup is more cost-effective than cluster systems needed to run larger LLMs. This highlights that the LLMs tested have greater access for universities and other organizations that have a wide range of available funds.

We must note that the experiments were run in mid-June to early July 2025. Since the field of multi-modal LLMs is ever changing and updating, it is possible other OCR capable LLMs are released after the submission of this article.

IV. Results

We did notice that some OCR technologies had a CER above 1, some very high (up to 60). While we were tempted to label these instances as outliers, we realized these instances were cases where the OCR technology had large errors. Since we want to mimic the needs of a real-world situation, we include these cases in the final results and provide discussion where such large errors skew the overall results. This situation highlights the important aspect of an OCR technology that is consistent in performance. For example, OCR technology *A* may have better CER scores overall compared to OCR technology *B* but *A* has several cases of large CER scores and *B* has none. In the real-world, one may want to sacrifice a little accuracy to avoid potential random large errors. We must also note that the LLMs did take considerably more time to run, e.g., approximately 17 or more minutes, whereas the non-LLM OCR technologies finished within 8 minutes.

For the overall results (Tables III and IV), we report the average for all metrics and group non-LLM and LLM OCR technologies. The best score for non-LLM and LLM OCR technologies are displayed in bold. Also, for the overall results, we report the Top CER % for each of the OCR technologies. Table III reports the CER, NLD, and BLUE scores, and Table IV reports the ROUGE scores. Among non-LLM-based OCR technologies, surya performed the best across all metrics. Among LLM-based vision OCR technologies, olmOCR and Mistral competed very closely where olmOCR having the better scores of the two except on average BLEU, average NLD, and Top CER %. We must note that Mistral had two clips with high CER, hence, the scores from these two clips skewed the average CER and also the average ROUGE scores. This was verified through excluding these two clips and reviewing the re-calculated results.

In this paper, we focus on the challenges that are specific to the nature of our documents, such as *faded*, *skewed*, *inverted*, and *special characters*. The *not* cases, e.g., *not skewed*, are also included for completeness. The results for these labels are seen in Table V. The results for clips that are both *faded* and *skewed*, are not included due to space constraints. Keep in mind that the average CER scores for Mistral are skewed due to the large CER error discussed earlier.

TABLE III: Comparing overall scores: CER, BLEU, and NLD. (Note that lower CER and NLD is better and higher BLEU is better. See Section II).

OCR	Avg. BLEU	Avg. CER	Avg. NLD	Top CER %
pytesseract	0.6765	0.2545	0.2467	0.4791
docTR	0.8126	0.1292	0.1278	0.5766
surya	**0.9089**	**0.0982**	**0.0467**	**0.8329**
Llama 3.2	0.9398	0.0326	0.0325	0.8106
olmOCR	0.9508	**0.0156**	0.0154	0.8747
Mistral	**0.9623**	0.1945	**0.0151**	**0.9053**
Pixtral	0.9016	0.2812	0.0522	0.7521
Llava 1.6	0.7627	1.2868	0.1823	0.3454
Phi-4	0.9363	0.2746	0.0350	0.8217

TABLE IV: Comparing overall scores: ROUGE. (Note that higher ROUGE is better. See Section II).

OCR	Avg. ROUGE-L	Avg. ROUGE-1	Avg. ROUGE-2
pytesseract	0.7534	0.7644	0.7422
docTR	0.8832	0.9323	0.8737
surya	**0.9536**	**0.9546**	**0.9413**
Llama 3.2	0.9668	0.9674	0.9591
olmOCR	**0.9853**	**0.9860**	**0.9795**
Mistral	0.9848	0.9852	0.9781
Pixtral	0.9463	0.9494	0.9312
Llava 1.6	0.8585	0.8635	0.8391
Phi-4	0.9641	0.9650	0.9538

Table V shows the performance of non-LLM and LLM OCR technologies across different labels. Our findings are:

1) *Faded*: The non-LLM-based OCR technologies pytesseract had the best average CER while surya had the highest Top CER %. For the LLM-based vision OCR technologies, olmOCR had the best average CER and Mistral the best Top CER %.
2) *Skewed*: Among the non-LLM-based OCR technologies, surya performs best. For the LLM-based OCR technologies, olmOCR scored best for average CER where Mistral comes ahead for the Top CER %.
3) *Inverted*: Among the non-LLM-based OCR technologies, surya performs the best. For the LLM-based OCR technologies, Mistral performs the best.
4) *Special Characters*: Among the non-LLM-based OCR technologies, docTR has the best average CER while Surya has the best Top CER %. For the LLM-based OCR technologies, olmOCR performs best.

Overall, among the OCR technologies tested, Mistral and olmOCR performed the best in all metric categories. However, olmOCR is the most consistent without any large errors (large CER scores).

V. Discussion

Each of the metrics chosen helps look at the results from different qualitative angles. CER and NLD scores can be used to identify minimal errors in OCR transcripts, while ROUGE and BLEU metrics further strengthen our hypothesis of LLM-based OCR to emphasize the quality of the resulting

TABLE V: Comparing CER and Top CER % scores for the *faded*, *skewed*, *inverted*, and *special characters* (SC) label states.

	Faded State				Skewed State				Inverted State				SC State			
	Avg. CER		Top CER %		Avg. CER		Top CER %		Avg. CER		Top CER %		Avg. CER		Top CER %	
OCR	faded	not	faded	not	skewed	not	skewed	not	inverted	not	inverted	not	S.C	None	S.C	None
pytesseract	**0.0869**	0.3335	0.6696	0.3893	0.3721	**0.0616**	0.3318	0.7206	0.2865	0.2320	0.4730	0.4834	0.3476	0.2510	0.0000	0.4971
docTR	0.1436	0.1225	0.6087	0.5615	0.1382	0.1145	0.5247	0.6618	0.1689	0.1014	0.5743	0.5782	**0.1598**	0.1281	0.0000	0.5983
surya	0.2450	**0.0291**	**0.7652**	**0.8648**	**0.0347**	0.2024	**0.8700**	**0.7721**	**0.1328**	**0.0740**	**0.8311**	**0.8341**	0.2148	**0.0939**	**0.1538**	**0.8584**
Llama 3.2	0.0160	0.0404	0.8696	0.7828	0.0456	0.0113	0.7892	0.8456	0.0387	0.0283	0.8108	0.8104	0.0662	0.0313	0.2308	0.8324
olmOCR	**0.0066**	0.0199	0.9043	0.8607	**0.0216**	**0.0057**	0.8655	0.8897	0.0170	**0.0146**	0.8851	0.8673	**0.0267**	**0.0152**	**0.5385**	0.8873
Mistral	0.5718	**0.0167**	**0.9304**	**0.8934**	0.3031	0.0164	**0.9058**	**0.9044**	**0.0076**	0.3256	**0.9527**	**0.8720**	0.1492	0.1962	0.1538	**0.9335**
Pixtral	0.1657	0.3356	0.6696	0.7910	0.4164	0.0595	0.7848	0.6985	0.1359	0.3831	0.7635	0.7441	0.1097	0.2876	0.1538	0.7746
Llava 1.6	1.0085	1.4179	0.3391	0.3484	1.1392	1.5288	0.3498	0.3382	0.9820	1.5006	0.3243	0.3602	10.0403	0.9579	0.0769	0.3555
Phi-4	0.0550	0.3780	0.9043	0.7828	0.3838	0.0954	0.7758	0.8971	0.3101	0.2496	0.8108	0.8294	0.9567	0.2489	0.4615	0.8353

text extracted by each OCR technology through the nature of what they measure: the quality of the generated text against the ground truth. The Top CER % provides a view of the overall performance of the OCR technology in text extraction. Large CER values (high error) are the reason some OCR technologies fall behind in average CER, but, filtering out the clips with a large CER value can show the potential of the OCR technology. However, one must consider if it is worth the risk of having high errors occur in their application. The clips with large CER values can be more easily spotted as the common symptom is long strings of repeated text. If this situation can be detected and there are relatively few of them, those images can be processed separately with another OCR technology. If many images with this problem are detected, another OCR technology may need to be used.

The overall results seen in Tables III and IV show that LLMs, specifically olmOCR and Mistral, are outperforming the other OCR technologies in all metric categories, sometimes by a large margin. LLMs outperformed non-LLMs 100% of the time. On the whole, we would like to emphasize the CER and NLD metric results as these are generally seen as a focus for OCR performance, especially if CER is 2% or lower. For OCR technologies that are close in some metrics, this small difference could conceivably mean these OCR technologies perform very similarly, as each time OCR is performed, there is some stochastic variability, as noted in [37]. Given that olmOCR and Mistral performed the best, we will focus further discussion on comparing these two.

From our results, we observed that a major factor in LLM performance is that of large CER values, i.e., a CER over 1. This is likely to occur when the OCR results are longer than the reference transcription text. We observed this to occur for Mistral in two cases. Reviewing these two clips revealed no pattern that would explain this behavior. We did see that, in this case, the OCR output repeated itself until the max output buffer was reached. Further investigations will focus on mitigating repetition by adjusting hyperparameters. We also want to fine-tune LLMs with our curated dataset.

We focus our discussion and observations on the results for clips with challenging characteristics. However, for completeness, the cases when clips are without these characteristics are presented in Table V. As seen in Table V, olmOCR and Mistral competed well against each other and best among the nine OCR technologies tested. Therefore, we can conclude that olmOCR and Mistral are more adept at handling documents with challenging characteristics. Despite olmOCR having fewer parameters versus Mistral (7B versus 24B, respectively), olmOCR remained competitive. We believe this is due to the fact that olmOCR was fine-tuned for OCR including historical documents.

The *faded* and *skewed* categories were handled well by both LLMs to varying degrees. This is promising, as arguably, these two categories may account for most situations found when performing OCR on scanned pre-digital historical documents. Of course, this is dependent on the nature of the documents and scans. Both LLMs also performed well for *inverted* clips. However, *special characters* posed more of a challenge. But we believe that fine-tuning on our dataset could minimize this.

We did observe that olmOCR was more efficient in terms of the time taken to complete the task, while Mistral took longer. Even though the Mistral model had more parameters (24B vs 7B), olmOCR had comparable performance. A challenge that can arise is the way models are trained and what they are expected to perform. olmOCR was designed to convert text from PDFs into machine-readable form, whereas Mistral vision helps users with image understanding. In other words, each was trained with specific data to help with an intended task. Despite this, interestingly enough, both performed very well on our task, even when no pre-processing was performed on the clips to "clean-up" visual characteristics.

For the non-LLM-based OCR technologies, Surya is recommended if resources to run LLMs are unavailable. One aspect to note is that docTR is very flexible with the different models to choose from for text detection and recognition, allowing the user to tailor docTR to their use case. Hence, choosing different models for text detection and recognition could provide better results than currently seen in our experiments. However, if resources permit, olmOCR is the preferred LLM from those we tested due to its consistency in low CER as opposed to using Mistral.

As a final discussion point, we performed OCR with a much larger openly available LLM with vision, Meta's Llama 4 Scout (`Llama-4-Scout-17B-16E-Instruct`) [38], which has 17B active parameters and 109B total parameters.

TABLE VI: Comparing overall scores of Mistral and olmOCR to Llama 4 Scout.

OCR	Avg. BLEU	Avg. CER	Avg. NLD	Top CER %
olmOCR	0.9508	0.0156	0.0154	0.8747
Mistral	0.9623	0.1945	0.0151	0.9053
Scout	**0.9688**	**0.0095**	**0.0089**	**0.9192**

The total amount of GPU RAM needed just to load this model was over 210 GB, substantially more than the other LLMs we tested. Llama 4 Scout did perform better than Mistral and olmOCR, but only by a small margin as seen in Table VI. This slight performance difference shows support that just using a larger model does not necessarily produce a substantial improvement.

VI. Text Analysis

Our experiments with our dataset (Section III-A) shows Mistral and olmOCR to perform the best among the OCR technologies. We choose olmOCR for future experiments as it performs consistently well. We were able to detect Mistral large CER error because of our transcribed dataset. However, this is not feasible as most real-world applications don't have a transcription or ground truth available for evaluating the quality of the extracted text. Hence, we use olmOCR to extract the text from all the real estate documents (or documents). We have 63 documents with 176 images (one page per image) in total, with each document comprising anywhere from 1 to 13 images. olmOCR took 88 minutes to extract text from all 63 documents' 176 pages.

We now use the extracted text from the documents for analysis. For our preliminary experiments, we started by grouping documents into bins of the decade based on the year the document was written. We then compare documents from one decade with all documents in other decades. For example, a document written in the 1920s was compared to all the other documents from the 1930s and 1940s. To compare the text, we choose an LLM that is known to perform well on reasoning based on HuggingFace's leaderboard [39]. Based on the leaderboard, we choose Qwen-3 model [40], a 32B parameter model capable of complex reasoning and a support for up to 32K context window. We prompt `Qwen3-32B` [41] model to compare documents. Our prompt instructed the model to "Analyze the texts provided and find the differences between them". Alongside the basic prompt, we send instructions to the model to guide it to output a response format that is computationally easy to parse. We ask the model to respond in the following format

1) **Difference in Text 1**: Highlight all the notable differences in Text 1.
2) **Difference in Text 2**: Highlight all the notable differences in Text 2.
3) **Summary of key differences**: Provide a summary of differences in bullet format.

It is important to note that Qwen3, with 32B parameters, needed a GPU with greater than 120 GB of memory for loading and inference. The response output consistently had a "summary of key differences" section for each comparison. We manually inspected multiple comparison results to make sure that the analysis was based on facts from the original documents and that the model did not hallucinate.

Our analysis revealed the following. Focusing on changes over time, substantial and nuanced differences were detected. The analysis showed that the legal agreements became more restrictive based on race over time, which is a meaningful finding when researching how race is perceived for housing restrictions. Another difference was the exclusion of spouses from the 1920s to the 1940s, but the inclusion of heirs in later years. The LLM reported changes in the structure and word changes, but in at least one case, manual inspection of the two compared documents did not reveal any noticeable difference. To fully verify this difference, one would need to closely read each document. These examples of identified differences are just the beginning to understanding this corpus. Through this process, one can gain insight into legal agreements that were purposefully racially restrictive, and further investigate this time period in history, a matter not as easily done otherwise.

VII. Conclusion and Future Work

In this paper, we explored the use of openly available LLMs with vision to perform OCR on scanned, pre-digital, historical documents from the early twentieth century, a new dataset we curated. We selected several other open-source non-LLM OCR technologies for comparison. Our results show that LLMs with vision can perform OCR well on document scans that have challenging characteristics, namely *fading*, *skewing*, text/background contrast *inversion*, and unique *special characters*. Two of the six LLMs, Mistral and olmOCR, performed the best in all categories. Hence, LLMs with vision are a feasible solution for some OCR tasks. Since OCR was performed on the original scans with no pre-processing, these LLMs are a great option for document preservation, especially for documents that pose specific challenges due to their age. The ability to perform OCR on the original scan without any image pre-processing is very beneficial as one may not know what corrections need to be applied to the images ahead of time. Alongside preservation, text extraction from challenging pre-digital historical documents can be used for additional analysis to uncover buried knowledge from these documents.

For future work, we would like to use the current curated dataset to aid in fine-tuning LLMs with vision to further improve their performance. We would like to explore quantization methods such as vLLM [42] to help with memory efficiency without compromising precision. Our current text analysis shows that substantial differences can be detected when using LLMs for analyzing the text; however, this is just the beginning. Additional automatic analysis of the extracted text can also help users understand the corpus, such as performing keyword extraction, classification, and content summarization.

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